

Daksh Sawhney

Email: dakshsawhney2008@gmail.com

Contact number: 9663745114

OBJECTIVE

High school student specializing in computational mathematics and quantitative finance. I translate abstract problems into structured models and test them in real environments.

Built Persifolio, an AI-based investment app and market simulator used by 4,000+ young professionals to learn budgeting, portfolio construction, and risk management. Published research on digital payment adoption in the *International Journal of Social Science and Economic Research*(IJSSER). Completing AS and A Level Further Mathematics in a single year.

Outside academics: competitive athlete (basketball, shot put) and environmental entrepreneur. Raised corporate funding to plant 100,000+ trees across six sites in Bangalore.

Interested in algorithmic modelling, quantitative analysis, and applying mathematical rigour to financial instability and systemic risk.

My three qualities: Curious, Enterprising, collaborative and Persistent

EDUCATION

Inventure Academy, Bangalore, India

2026 – 2027

- Cambridge A Level Grade 12
- Mathematics, Further Mathematics, Physics, Chemistry, Computer Science, English

Inventure Academy Bangalore, India

2025 – 2026

- Cambridge AS Level Grade 11 (grade A)  AS maths.pdf
- Mathematics (**grade A**), Physics, Chemistry, Computer Science

Inventure Academy, Bangalore, India

2024- 2025

- *International General Certificate of Secondary Education IGCSE*
English Language(A), English Literature(A*), Mathematics(A*), Additional Mathematics(A*),
Biology(A*), Physics(A*), Chemistry(A*), Economics(A*), Computer Science(A*), Spanish (A)
-  Daksh transcript 2024-2025.pdf

STANDARDISED TESTING

Name of the Exam	Date	Details
SAT		Attempting in August 2026
IELTS / TOEFL		
CEMC/ OTHER TEST		

ACADEMIC HONORS AND AWARDS

Crest Gold : Persifolio: Personalised Portfolio; Virtual Stock Investment Simulator	November 2025	awards (1).pdf
Harvard Hackathon: NATIONAL WINNER: Designed and built a web application in 48 hours with a 5-member interdisciplinary team. It is a web app, a research-driven prototype developed to investigate how artificial intelligence, real-world job market data, and document-centric interface design can be combined to support structured career preparation. The system reframes the résumé not as a template produced at the end of education, but as a continuously evolving representation of progress that is directly coupled with skill acquisition, project work, and experiential learning.	Jan 2026	 hack harvar...
Cambridge International Certificate of Education-Distinction	June 2025	ICE CERTIFIC ATE.pdf
Inventure Big Leap Award: English Literature	2023-24	school awards.pdf

OTHER AWARDS

Inventure Academy Award of Excellence:: Community Outreach Award	2024-25	school awards.pdf
Inventure Academy Award of Excellence : Inventure Sports Award	2024-25	school awards.pdf sports awards.pdf
Inventure Academy Award of Excellence : Inventure Sports Award	2024-25	school awards.pdf sports awards.pdf
Inventure Academy Award of Excellence : Whiz kid	2025-26	comouter whiz award.pdf
Inventure Academy Award of Excellence : Inventure Scholar Award	2025-26	scholar award.pdf
Inventure Academy Award of Excellence : Computer Science	2025-26	computer subject topoer.pdf
Inventure Academy Award of Excellence: Multiachiever award	2025-26	multi achiever award.pdf
Portfolio Strategy Challenge -Winner, HDFC Credelia stock portfolio designed for a client and won First place in the investment pitch competition.	Jan-Feb 2024	HDFC challenge.pdf

EXTRA-CURRICULAR ACTIVITIES*(Participation counts too. Please add more rows as required)*

Name of the Activity	Location / Level	Achievement	Grade Month / Year	Links : Certificate / Website / Blog/ Photos / Newspaper Publication	Staff's name and Email id who can verify your activity
TISB swimming Bronze Medal: 50m Butterfly (Under-15)	Inter- School	3rd Prize	Grade 9 15 th June 2023	TISB aquatic meet.pdf	Mr Fredy
INMUN BBMP committee	Interschool	Participation	Grade 9 2023		
ISSO Badminton	National	Participation	2024	photos	Mr Fredy
Head start interschool basketball	Interschool	Silver medal	Grade 9, 2024	sports awards.pdf	
Interschool athletic relay	Interschool	Bronze	2024	sports awards.pdf	Mr.Dinesh
Inventure Shotput throw	Interschool	Gold	2024	sports awards.pdf	Mr.Dinesh
Inventure Shotput throw	Interschool	Silver	2025	shotput second.pdf	Mr Dinesh
ISSO Shotput throw	National	Participation	2025	Daksh isso shotput.pdf	
School Club: Tech for change	School		2024-2026	Created Pathforge and trying to build an e-ballot platform for school elections	

Online Courses / Subject-specific pursuits

1. Name of the Course: Technical Analysis in Trading and Stock Market

University/ Organisation: Hexarum Co

Grade: 9

Date/ Duration: April 2024 – June 2024

Details: Completed technical analysis certification (Hexarum) covering candlestick charting, support/resistance identification, pattern recognition, and momentum indicators (Moving Averages, RSI, MACD) for equity market analysis.

Link to the Certificate: [hexaurum.pdf](#)

2. Name of the Course: Introduction to Psychology

University/ Organisation: Coursera/Yale University

Grade: 10

Date/ Duration: August 2023

Details: Completed foundational psychology coursework covering decision-making frameworks, risk perception, behavioral economics, and cognitive biases in competitive and financial contexts.

Link to Certificate [Psychology Yale.pdf](#)

3. Name of the Course: Certificate Course in Introduction to Data Science & AI

University/ Organisation: IIT Madras

Grade: 11

Date/ Duration: Oct 2025-December 2025

Details: Cleaned and analyzed real e-commerce transaction data; applied data extraction, preprocessing, and visualization techniques to identify purchasing patterns and customer behavior trends using Python-based data science workflows.

Link to Certificate: [IIT madras.jpg](#)

4. Name of the Course: FinTech: Foundations, Payments, and Regulations

University/ Organisation: Coursera/University of Pennsylvania

Grade: 11

Date/ Duration: December 2025

Details: Completed fintech certification covering digital payment systems, peer-to-peer lending platforms, robo-advisory algorithms, regulatory technology (RegTech), and mobile banking infrastructure.

Link to Certificate: [Finetch foundtion.pdf](#)

5. Name of the Course: Cryptocurrency and Blockchain: An Introduction to Digital Currencies

University/ Organisation: Coursera/University of Pennsylvania

Grade: 11

Date/ Duration: December 2025

Details: Completed blockchain technology certification covering distributed ledger architecture, consensus mechanisms, smart contract development, and cryptocurrency transaction validation protocols.

Link to Certificate: [crypto currency.pdf](#)

6. Name of the Course: Lending, Crowdfunding, and Modern Investing

University/ Organisation: Coursera/University of Pennsylvania

Grade: 11

Date/ Duration: December 2025, 1 month

Details: Studied democratized finance platforms: robo-advisors, marketplace lending, and crowdfunding infrastructure. Applied Modern Portfolio Theory to algorithmic asset allocation models and analyzed peer-to-peer lending networks that bypass traditional banking intermediaries. Evaluated credit risk assessment using alternative data sources through real-world case studies.

Link to Certificate [crowdfunding.pdf](#)

7. Name of the Course: Application of AI, InsurTech, and Real Estate Technology

University/ Organisation: Coursera/University of Pennsylvania

Grade: 11

Date/ Duration: December 2025, 1 month

Details: Explored AI applications in insurance claims automation, customer personalization, and real estate pricing optimization. Studied how machine learning models improve property valuation accuracy, accelerate transaction decisions, and enhance long-term investment performance.

Link to Certificate: [Coursera | Online Courses & Credentials From Top Educators. Join for Free | Coursera](#)

8. Name of the Course: Coursera Fintech: Foundations & Applications of Financial Technology Specialisation

University/ Organisation: Coursera/ University of Pennsylvania

Grade: 11

Date/ Duration: December 2025

Link to Certificate: [Fintech U penn.pdf](#)

9. Name of the Course: Moneyball: AI, Sports, and Business project

University/ Organisation: Inspirit AI, Stanford University graduates, USA

Grade: 10

Date/ Duration: May 2024-June 2024

Details: Applied Moneyball methodology to baseball player valuation: used machine learning clustering algorithms to identify undervalued players within budget constraints and project replacements for aging or departing roster talent.

Performed statistical analysis on historical player performance data to isolate metrics with highest predictive power for success (OBP, SLG, WAR derivatives). Built clustering models to group players by performance profile and flag budget-efficient alternatives to high-cost veterans.

Link to Certificate: [Inspirit AI Scholars Certificate.pdf](#)

10. Name of the Course: Commerce Club

University/ Organisation: Inventure Academy/ Hari Sir

Grade: 9,10

Date/ Duration: June 2023 to Oct 2023

Details: Published an article in the Commerce Club magazine. THE ECONOMIC SHIFT, The Invisible Hand, Feb 2024, Issue 1.

Link to Certificate: [Commerce club magazine.pdf](#)

Research Experience

1. RESEARCH SCIENCE INITIATIVE-INDIA- FROM JUNE 2026 TO JULY 2026

Selected for Research Science Institute (RSI-India), a MIT-affiliated summer research program with <2% acceptance (30 students nationally). Conducted research under Dr. Kaushal Verma (Dean of Mathematics, Indian Institute of Science) on transient chaos and geometric transitions in nonlinear dynamical systems.

Designed and coded a custom bifurcation diagram simulator to analyze how discrete-time nonlinear equations transition into chaos and to quantify predictability of long-term behavior. Developed a structural classification coefficient (ξ) that categorizes equations into smooth unimodal, smooth multimodal, and piecewise linear classes based on algebraic form, enabling prediction of dynamical behavior directly from equation structure.

Manuscript is submitted to the journal and is under review

[rsi certificate.pdf](#)

[IMG_3523.PNG](#)

2. Cambridge Centre of International Research, *Cambridge, UK*

From Nov 2025-March 2026.

UNDER THE MENTORSHIP OF DR WEI WEI SUN, LECTURER, DEPT OF COMPUTER SCIENCE, UNIV OF CAMBRIDGE, UK

Summary: The paper investigated the types of sentence patterns that lead to tagging errors in LLM models because of the use of informal Hindi and English language. The error analysis revealed that the most frequent misclassifications occurred between nouns, verbs, proper nouns, and adjectives.

STATUS OF PAPER: Submitted for an international conference in artificial intelligence and is under review.

[CCIR certificates.pdf](#)

[POS Tagging of Hindi English Code Mixed Text Using an Averaged Perceptron Draft_DakshSawhney \(4\).pdf](#)

3. Publication of research paper Nov 2024-Jan 2024

From Cash to Clicks: The Transformative Impact of UPI on Consumer Behaviour and Financial Inclusion in India: International Journal of Social Science and Economics, <https://doi.org/10.46609/IJSSER.2025.v10i07.021>

Summary: This paper investigates the transformative impact of the Unified Payments Interface (UPI) on consumer behaviour, financial inclusion, and economic growth in India

Internships

This section highlights the Internships: This includes paid or unpaid, part-time/full-time, Online/ in-person

Summer Intern In person	Nine leaps Technology Solutions Ltd AI Centre of Excellence at Noviro.ai	1st April to 15th May 2025	Gained hands-on experience in web development, learning frontend (HTML/CSS) and (Python-Flask) integration. Developed proficiency in databases, including SQL and SQLite, performing CRUD operations within web applications. Integrated Chatbots and Large Language Model (LLM) APIs, building conversational AI tools. Explored Retrieval-Augmented Generation (RAG) architectures, vector embeddings, and their applications to enhance AI responses.	Internship Completion Certificate Daksh Sawhney.pdf Letter of Recommendation.pdf	Mr. Anand anand.ganesan@nineleaps.com
Software development intern	Address Makers Pvt Ltd	From May 2025 to October 2025	Designed and developed Gen AI chatbot and intelligent automation bot for the company. This was not a prototype, but rather a production-grade system that the company actively uses for customer interaction. The solution integrated multiple linguistic language models to serve users across regional language preferences and incorporated speech-to-text technology to enable voice-based customer interaction.	address maker lor.pdf LOR from company	Mr Santosh Soni

VOLUNTEER SERVICE / COMMUNITY OUTREACH / ADDITIONAL INFORMATION

(This section highlights your bit towards society. Activities, Raising of Funds, Drives, etc. you have done for your society like: Interned at an NGO, Organized events like sports day, etc. for underprivileged children, Service Projects)

Name of the activity	Location	Grade Date/ Duration	Details	
Shades of Tomorrow — Miyawaki Forest Project	Bangalore	9th grade onwards	Raised ₹500,000 in corporate funding by presenting impact case studies and demonstrating technical expertise in urban reforestation. Led seven plantation drives across Bangalore, planting 100,000+ trees in partnership with Indus International School and corporate sponsors.	Tree plantation drives LOR from the company india build plantation.pdf
RGH Government School	Bangalore	June 2024- Nov 2024	Tutored 5 students in English for 2 years, helping them achieve 25 per cent higher marks in exams through one-on-one tutoring and monthly practice sheets. Developed adapted learning materials, visual aids, and activities to support students with academic gaps in Grades 6 and 8 using design thinking processes.	Ms Lakshmi/Ms Nikitha
Empower-Fin Project	Tribes for good	1May 2024-1st June 2024	Addressed Financial Management through multiple awareness workshops on economics concepts such as inflation and shrinkflation for young students and women from economically weaker sections. Conducted finance workshops for peers at my school and neighbourhood. Completing 25 hours of community work.	Jinal Rajpopat < jinal@tribesforgood.com > https://youtu.be/zuuYRyHGc7M?si=bG5IUeEZEXshnJLz Ecofin.pdf

--	--	--	--	--

LEADERSHIP ROLES

(Mention all the Leadership tasks/responsibilities that were handed to you. Examples are mentioned below. You could also mention examples of how various situations brought out the Leadership Quality in you)

Sr No	Roles	Grade Month Year
	Founder, creator and Social entrepreneur Persifolio: AI-based Investment Portfolio Recommendation system with market stimulation. Persifolio App has cleared the first stage of patenting and has been published in the Indian Patent Journal, recently, it was adopted by schools in Odisha as part of their financial awareness program, supported by a Member of the Rajya Sabha, Govt of India Built Persifolio, a mobile investment education platform bridging the wealth gap for underserved communities. Conducted workshops and onboarded 4,000+ young adults and low-to-mid income users, teaching budgeting, portfolio construction, and risk management through hands-on simulation. The app simplifies complex financial concepts with personalized investment suggestions, risk scores, and a virtual stock market simulator. Reduces barriers to entry by addressing fear and misinformation around investing. Users learn to build financial independence and break cycles of debt. Technical implementation: Android app built with Flutter and Firebase. Integrated APIs for real-time market data, Material 3 UI, Crashlytics for stability monitoring, and Google Analytics for user behavior tracking. Published on Google Play. Algorithm generates personalized portfolios and risk assessments based on user-inputted goals, interests, and risk appetite.	<i>From 2024 onwards</i> patent certificate.pdf awards (1).pdf persifolio letter.pdf LOR for Daksh Sawhney by MP.pdf Images.zip
	Creator Cybersecurity Project: Cyberslayer: a web app for Cybersecurity Awareness for Senior Citizens of India Motto: Don't be a victim, be a slayer.	Sept 2025 onwards

	Built a free, quiz-based cybersecurity awareness platform protecting senior citizens from scams and fraud. Simulates real-world threats (fake arrest scams, phishing emails, impersonation attacks) through interactive scenarios, teaching users to recognize, respond to, and report digital exploitation. Addresses the human vulnerability gap in cybersecurity. Seniors, small business owners, and non-tech-savvy populations are targeted through deception and fear. The platform turns passive advice into active practice, reducing financial and psychological damage from fraud.	
--	--	--

OTHER INTERESTS AND PURSUITS

(This section highlights your Hobbies other than the ones mentioned above. Examples are below)

Sr No	Activity	Month and Year
	https://medium.com/@dakshsawhney2008/does-the-profit-motive-bring-out-the-best-or-the-worst-in-us-bb48159c1bbd Description: 100 words Links: Website/Blog / Photos / Newspaper Publication (If Any)	February, 2026
	Hiking (Hiked to Gomukh Himalayas, 14,000 feet)	2023
	Badminton (won gold in mixed adult doubles in Bangalore) Inventure MUN 2023- BBMP committee Agenda- 'Impact of unplanned urbanisation in the Whitefield area'	2023
	Coding skills: AI/ML Models: OpenAI GPT, Google Gemini, Cursor, Python, WebDev ● Cloud Platforms: AWS, Azure ● Databases: MongoDB, Firebase ● Programming: Python, SQL, Kaggle	